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*Exploration and Extension in Spatial Extremes Incorporating
Thin-plate Splines Basis Functions*



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USA EPA AQS CMAQ CMAQ Morris skew-t process matern
covariance modeling gaussian noise term CMAQ Morris time-varying coeffi-
cients with AR(1) structure, censoring to the tail, partition time-varying
coefficients expand to AR(2) structure, and incorporate multiple resolution thin-plate
splines basis functions as covariates to improve modeling non-stationarity spatial de-
pendence.

Abstract

To more accurately calibrate CMAQ by the ozone site data, Morris develop partition, threshold, AR(1) time-varying coefficients based on spatial skew-t process. To let model more scalable, we extend the time-varying coefficients to AR(2) structure. And incorporate the multiple thin-plate spline basis functions as covariates to capture the non-stationary spatial dependence. We find that in some level of exceedance, the model with the AR(2) time varying coefficients or the mrts covariates have a lower brier score than the Morris model. Give user more choice in model selection.

Keywords:

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1 Introduction

2 Methodology

2.1 Spatio-temporal skew-t process

The methodology in this thesis builds on the spatio-temporal skew- t model of Morris et al. (2017). We briefly summarize that model here so that the AR(2) and MRTS extensions in the next two subsections can be stated against a common notation.

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \dots, n_t\}$. Following Morris et al. (2017), we model

$$Y_t(\mathbf{s}) = \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}), \quad (1)$$

where $\mathbf{X}_t(\mathbf{s})$ is a p -vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic Matérn correlation

$$\rho(h) = (1 - \gamma) I(h = 0) + \gamma \frac{1}{\Gamma(\nu) 2^{\nu-1}} \left(\sqrt{2\nu} \frac{h}{\varphi} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{h}{\varphi} \right), \quad (2)$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\gamma \in [0, 1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \dots, Y_t(\mathbf{s}_n)]^\top$ follows an n -dimensional skew- t distribution; consequently (??) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t , let $\mathbf{w}_{t1}, \dots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} = \{\mathbf{s} \in \mathcal{D} : k = \arg \min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \quad k = 1, \dots, K. \quad (3)$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \stackrel{iid}{\sim} N(0, 1)$ and $\sigma_{tk}^2 \stackrel{iid}{\sim} \text{IG}(a/2, b/2)$. Within each subregion the model retains the skew- t form, while sites

in different partitions are not asymptotically dependent.

Because the EPA compliance criterion focuses on exceedances of a high level L , the model fits only the upper tail by censoring the data at a pre-specified threshold $T \leq L$. Writing the partially censored response $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\tilde{\mathbf{Y}}_t = [\tilde{Y}_t(\mathbf{s}_1), \dots, \tilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk} , and the scale parameters σ_{tk}^2 . To preserve the marginal skew- t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform:

$$w_{tki}^* = \Phi^{-1} \left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)} \right], \quad i = 1, 2, \quad (4)$$

$$z_{tk}^* = \Phi^{-1} \{ \text{HN}(z_{tk}) \}, \quad (5)$$

$$\sigma_{tk}^{2*} = \Phi^{-1} \{ \text{IG}(\sigma_{tk}^2) \}, \quad (6)$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}_{tk}^*, z_{tk}^*, \sigma_{tk}^{2*}$ has a standard normal marginal distribution, and Morris et al. (2017) place a stationary AR(1) Gaussian time series on each of them:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^* \sim N_2(\phi_w \mathbf{w}_{t-1,k}^*, (1 - \phi_w^2) \mathbf{I}_2), \quad (7)$$

$$z_{tk}^* \mid z_{t-1,k}^* \sim N(\phi_z z_{t-1,k}^*, 1 - \phi_z^2), \quad (8)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*} \sim N(\phi_\sigma \sigma_{t-1,k}^{2*}, 1 - \phi_\sigma^2), \quad (9)$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$, and initial values $\mathbf{w}_{1k}^* \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z_{1k}^* \sim N(0, 1)$, $\sigma_{1k}^{2*} \sim N(0, 1)$. Transforming back through (??)–(??) gives $\mathbf{w}_{tk} \sim \text{Unif}(\mathcal{D})$, $z_{tk} \sim \text{HN}(0, 1)$, and $\sigma_{tk}^2 \sim \text{IG}(a/2, b/2)$ at every t , so that the spatial skew- t structure of (??) is preserved at each day. The complete hierarchical model is fitted via MCMC with Gibbs and Metropolis–Hastings steps, with predictions at unobserved sites obtained by Bayesian kriging on the multivariate normal full conditional of $\mathbf{Y}_t$.

This skew- t baseline – partitioned, censored, and AR(1)-correlated in time – defines the reference specification against which the two proposed extensions of this thesis are evaluated.

2.2 Extension to AR(2) time-varying coefficients

The AR(1) time series in (??)–(??) links each transformed parameter on day t to its value on day $t - 1$ only. This may be too restrictive for daily atmospheric variables such as ozone, where persistence and weather-driven cycles can carry information across multiple lags. We therefore replace the AR(1) prior on each transformed parameter by a stationary AR(2) prior, while preserving the unit marginal variance required by the probability integral transforms in (??)–(??).

The AR(2) recursion. For a generic transformed parameter $\{Y_t\}_{t=1}^{n_t}$ – standing in for any of the scalar components of $\mathbf{w}_{tk}^*$, or for z_{tk}^* or σ_{tk}^{2*} at a fixed knot k – the AR(2) prior is

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2), \quad t \geq 3. \quad (10)$$

For the recursion to define a stationary process, (ϕ_1, ϕ_2) must lie inside the stationarity triangle

$$|\phi_2| < 1, \quad \phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1. \quad (11)$$

Just as in the AR(1) case, the AR(2) parameters $(\phi_1^{(w)}, \phi_2^{(w)})$, $(\phi_1^{(z)}, \phi_2^{(z)})$, and $(\phi_1^{(\sigma)}, \phi_2^{(\sigma)})$ are separately defined for $\mathbf{w}_{tk}^*$, z_{tk}^* , and σ_{tk}^{2*} .

Yule–Walker moments under unit marginal variance. We require $\gamma_0 := \text{Var}(Y_t) = 1$ so that the inverse PIT applied to Y_t recovers the half-normal, inverse-gamma, or uniform target margins on the original scale. Multiplying (??) by Y_{t-h} and taking expectations yields the Yule–Walker recursion

$$\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}. \quad (12)$$

Setting $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ gives

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad (13)$$

and multiplying (??) by Y_t pins down the innovation variance

$$\sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \quad (14)$$

Setting $\phi_2 = 0$ in (??)–(??) recovers the AR(1) form $\gamma_1 = \phi_1$, $\sigma_\epsilon^2 = 1 - \phi_1^2$, so the AR(2) prior strictly contains the AR(1) prior used by Morris et al. (2017).

Stationary joint of the initial pair. Because the recursion (??) requires two seeds, the joint distribution of (Y_1, Y_2) has to be specified separately. Independent $N(0, 1)$ initial values would create a transient regime in which $\text{Var}(Y_t) \neq 1$ for the first several days, breaking the PIT mechanism that produces the correct skew- t margin. Stationarity therefore forces the bivariate Gaussian initial condition

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \quad (15)$$

which yields the conditional density used when updating Y_2 alone:

$$Y_2 \mid Y_1 \sim N(\gamma_1 Y_1, 1 - \gamma_1^2). \quad (16)$$

Note that the mean of (??) is $\gamma_1 Y_1$, not $\phi_1 Y_1$, and its variance is $1 - \gamma_1^2$, not σ_ϵ^2 ; the two expressions coincide only when $\phi_2 = 0$. In the AR(2) implementation it is essential that both the simulator and the MCMC density at $t = 2$ use (??) so that the marginal property $\text{Var}(Y_t) = 1$ holds at every t .

Full conditional with three prior factors. The AR(2) Markov structure factorizes the joint prior of $(Y_1, \dots, Y_{n_t})$ as

$$p(Y_1, \dots, Y_{n_t}) = p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \quad (17)$$

with $p(Y_1, Y_2)$ given by (??) and each conditional factor for $s \geq 3$ following (??). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (??) through three factors: the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor

$p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of Y_t is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \quad (18)$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (??). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R = \min \left(1, \frac{\pi(Y_t^{(c)} \mid Y_{-t}, \mathbf{Y})}{\pi(Y_t \mid Y_{-t}, \mathbf{Y})} \right), \quad (19)$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (??).

Application to all three transformed parameters. The AR(2) prior (??) together with the stationary initial condition (??) replaces (??)–(??) for each transformed parameter:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^*, \mathbf{w}_{t-2,k}^* \sim N_2 \left(\phi_1^{(w)} \mathbf{w}_{t-1,k}^* + \phi_2^{(w)} \mathbf{w}_{t-2,k}^*, \sigma_{\epsilon,w}^2 \mathbf{I}_2 \right), \quad (20)$$

$$z_{tk}^* \mid z_{t-1,k}^*, z_{t-2,k}^* \sim N \left(\phi_1^{(z)} z_{t-1,k}^* + \phi_2^{(z)} z_{t-2,k}^*, \sigma_{\epsilon,z}^2 \right), \quad (21)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*}, \sigma_{t-2,k}^{2*} \sim N \left(\phi_1^{(\sigma)} \sigma_{t-1,k}^{2*} + \phi_2^{(\sigma)} \sigma_{t-2,k}^{2*}, \sigma_{\epsilon,\sigma}^2 \right), \quad (22)$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies inside the stationarity triangle (??), the innovation variances are obtained from (??)–(??), and the initial pairs $(\mathbf{w}_{1k}^*, \mathbf{w}_{2k}^*)$, (z_{1k}^*, z_{2k}^*) , $(\sigma_{1k}^{2*}, \sigma_{2k}^{2*})$ follow the bivariate Gaussian (??) component-wise. After inverse transformation through (??)–(??), the back-transformed parameters $\mathbf{w}_{tk}$, z_{tk} , σ_{tk}^2 retain the same marginal distributions as in the AR(1) baseline, so that the spatial skew- t model (??) is preserved at every t . Setting $\phi_2^{(\cdot)} = 0$ in (??)–(??) recovers the Morris AR(1) specification, so the AR(2) extension nests the original time-series model and adds one additional autoregressive coefficient per latent process.

2.3 Incorporate MRTS covariates

3 Simulation study

3.1 Design

The simulation study is used to evaluate whether the proposed extensions improve predictive performance beyond the space-time skew- t threshold model of Morris et al. (2017). The benchmark simulation follows the structure of Morris et al.: observations are generated at $n_s = 144$ spatial sites, sampled uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ independent time points. The mean structure is constant in the data generation, $\mathbf{X}^\top \boldsymbol{\beta} = 10$, while the fitted models include an intercept and first-order spatial trend. The main data-generating settings include Gaussian data, skew- t data with one knot, skew- t data with five knots, a max-stable setting, and a Brown-Resnick max-stable setting. For each setting, 50 simulated data sets are generated.

For the baseline comparison, five Bayesian models are considered:

1. Gaussian marginal distribution with $K = 1$ knot;
2. Skew- t marginal distribution with $K = 1$ knot and no thresholding;
3. Symmetric- t marginal distribution with $K = 1$ knot and threshold $T = q(0.80)$;
4. Skew- t marginal distribution with $K = 5$ knots and no thresholding;
5. Symmetric- t marginal distribution with $K = 5$ knots and threshold $T = q(0.80)$.

Here $q(0.80)$ denotes the empirical 80th percentile of the simulated response. These five models represent the main combinations of tail behavior, thresholding, and spatial partitioning used in the Morris model.

Two additional simulation experiments are conducted for the extensions in this thesis. First, to evaluate the AR(2) time-varying coefficient structure, data are generated with temporal dependence in the latent τ , $\mathbf{w}$, and $\mathbf{z}$ processes. The experiment uses $n = 100$ sites and $T = 30$ time points, with 90 sites used for training and 10 sites withheld for prediction. We repeat the experiment 20 times and compare three temporal structures: no time series, AR(1), and AR(2). Each fitted model uses a skew- t marginal distribution with $K = 5$ knots and no thresholding, and the MCMC is run for 5000 iterations with 2500 burn-in iterations.

A complementary AR(2) experiment embeds the temporal mechanism directly into the benchmark simulation, so that the AR(2) analysis model can be tested against data with a known autoregressive structure. Two further data-generating settings are added. Both produce one-knot skew- t data ($K = 1$, $\lambda = 3$) on the same $n_s = 144$ sites and $n_t = 50$ time points as the benchmark, but the latent τ , $\mathbf{w}$, and z processes now evolve through a fixed AR(2) recursion (??)–(??) whose coefficients are held constant across time. Setting 9 uses a strongly dependent coefficient pair $(\phi_1, \phi_2) = (0.80, -0.35)$, and Setting 10 a weakly dependent pair $(\phi_1, \phi_2) = (0.12, -0.05)$; both pairs lie inside the stationarity triangle (??). For each setting, 50 data sets are simulated and split into 100 training and 44 validation sites. Each data set is fit by seven analysis methods: the five baseline models listed above, together with two AR(2)-augmented skew- t models—one with $K = 1$ knot and one with $K = 5$ knots—whose time-varying coefficients follow the AR(2) prior (??)–(??). Because the data are themselves generated with AR(2) dependence, this comparison gives the temporal extension its most favorable test.

Second, to evaluate the MRTS covariate extension, we augment the original five models with multi-resolution thin-plate spline basis functions (Tzeng and Huang, 2018). These basis functions are added to the regression component in order to capture nonstationary spatial structure that may not be fully represented by a stationary Matérn covariance. We focus on simulation settings 4 and 5: setting 4 is a one-knot skew- t data-generating process with $\lambda = 3$, and setting 5 is a five-knot partitioned skew- t data-generating process with $\lambda = 3$. For setting 4, each original model is compared with versions using 10, 15, and 20 MRTS covariates. For setting 5, each original model is compared with a version using 15 MRTS covariates. Each setting uses 50 simulated data sets, and each MCMC chain is run for 20000 iterations with 10000 burn-in iterations.

3.2 Cross validation

Predictive performance is evaluated by withholding spatial sites from model fitting and scoring posterior predictive distributions at the held-out sites. In the benchmark simulation, 100 sites are used as training sites and 44 sites are withheld for validation. In the AR(2) experiment, the split is 90 training sites and 10 testing sites. The MRTS experiment follows the benchmark simulation structure and compares each MRTS-augmented model with the corresponding non-MRTS model on the same simulated data sets.

Because the main inferential target is prediction of high-level exceedances, the pri-

mary scoring rule is the Brier score (Gneiting and Raftery, 2007). For an exceedance level L , the Brier score is $\text{BS}(L) = \{I(y > L) - \hat{P}(Y > L)\}^2$, where y is the held-out observation and $\hat{P}(Y > L)$ is the posterior predictive probability of exceeding L . The score is averaged over all held-out sites and time points, and lower values indicate better exceedance prediction.

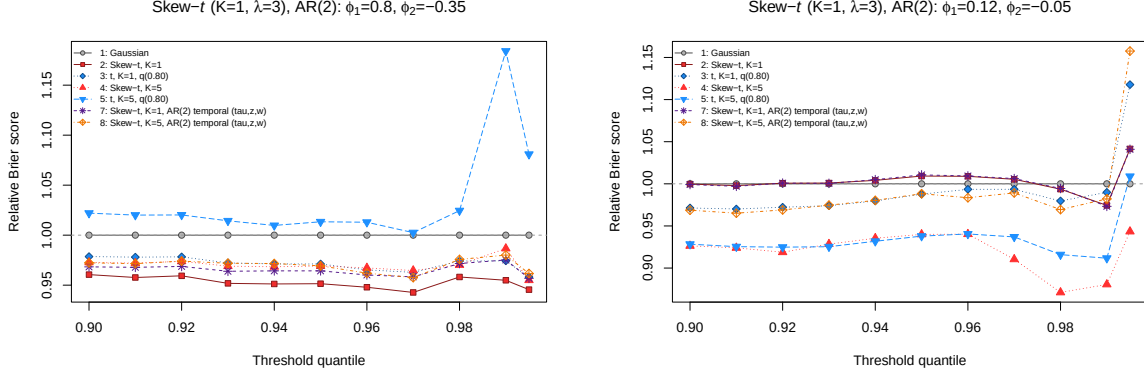
For the extension experiments, we also record MAPE and quantile scores. The quantile score at level α is used to evaluate the accuracy of posterior predictive quantiles, with lower values again indicating better prediction. The reported exceedance and quantile levels are concentrated in the upper tail: the AR(2) experiment uses $q(0.90)$, $q(0.95)$, $q(0.98)$, and $q(0.99)$, while the MRTS experiment uses levels from $q(0.90)$ to $q(0.995)$.

3.3 Results

The benchmark simulation supports the main conclusion of Morris et al. (2017): when the fitted model matches the data-generating mechanism, it tends to improve predictive performance, and when the data are non-Gaussian the skew- t models generally improve over a Gaussian model. Thresholded models are not uniformly better in the simulation setting, which is expected because most simulated data are generated directly from the full skew- t model rather than from a censored tail model. The simulation therefore confirms that the skew- t framework provides flexibility for asymmetric and heavy-tailed data while remaining competitive with Gaussian methods when the data are light-tailed and symmetric.

Figure ?? examines the AR(2) extension directly. All seven analysis methods are fitted to the two fixed-AR(2) data-generating settings, and for each method the Brier score is plotted relative to the Gaussian model across upper-tail threshold quantiles. A value below the dashed reference line at 1 means the method predicts exceedances better than the Gaussian baseline at that quantile, and a value above it means worse.

Under strong serial dependence (Setting 9, Figure ??), every skew- t and symmetric- t method except the five-knot thresholded t -model improves on the Gaussian baseline, with relative Brier scores between roughly 0.94 and 0.98. The one-knot skew- t model, which matches the $K = 1$ data-generating truth, is the best method, with a relative score near 0.95. Critically, adding the AR(2) temporal structure does not help: the one-knot AR(2) skew- t curve sits essentially on top of—and slightly above—the one-knot non-temporal skew- t curve, and the five-knot AR(2) model tracks its non-temporal



(a) Setting 9: strong AR(2) dependence, $(\phi_1, \phi_2) = (0.80, -0.35)$. (b) Setting 10: weak AR(2) dependence, $(\phi_1, \phi_2) = (0.12, -0.05)$.

Figure 1: Relative Brier score (versus the Gaussian model) by threshold quantile for the seven analysis methods fitted to the fixed-AR(2) data-generating settings. Values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline; values above it indicate worse.

counterpart. The five-knot thresholded t -model is the only method worse than Gaussian, and its relative score degrades sharply at the 0.99 quantile.

Under weak serial dependence (Setting 10, Figure ??) the pattern shifts. The five-knot skew- t and five-knot thresholded t models are now clearly best, with relative Brier scores mostly between about 0.88 and 0.94, while the one-knot models—including the one-knot AR(2) model—are essentially indistinguishable from the Gaussian baseline. Once again the AR(2) and non-temporal curves nearly coincide at each knot count. Far-tail estimates are noisy: at the 0.995 quantile several methods rise above 1 because few observations exceed such extreme thresholds.

Two messages emerge. First, on data that are explicitly generated with AR(2) dependence—the most favorable possible setting for the temporal extension—fitting an AR(2) analysis model yields no measurable Brier-score gain over the corresponding non-temporal skew- t model at any quantile. Predictive accuracy is governed by the marginal and spatial flexibility of the model, namely the skew- t margin and the number of knots, rather than by the autoregressive order of the latent coefficients. Second, the preferred knot count depends on the dependence regime: strong temporal dependence favors the parsimonious one-knot model that matches the generating truth, whereas weak dependence favors the richer five-knot partition. This reinforces the conclusion that the AR(2) structure should be treated as a candidate option rather than a default

improvement.

To integrate the proposed extensions in one comparison framework, we summarize AR(2) and MRTS evidence together. For AR(2), Table ?? reports selected quantitative results from the detailed missing-at-location experiments in `ar2_extend_detail_doc.md`. Across scenarios, AR(2) sometimes improves mid-tail Brier scores (for example BS90/BS95 in stronger-dependence settings), but AR(1) or no-time-series can still be preferable for runtime or at specific tail levels. This reinforces that temporal flexibility does not imply uniform predictive dominance.

Scenario (missing-at-location)	BS90 (AR2/AR1/NoTS)	BS95 (AR2/AR1/NoTS)	BS99 (AR2/AR1/NoTS)	Time(s) (AR2/AR1/NoTS)
σ^2 AR(2) only	0.0129/0.0130/0.0128	0.0106/0.0106/0.0106	0.0032/0.0032/0.0032	54.78/54.53/51.41
All large ϕ	0.0344/0.0346/0.0352	0.0269/0.0267/0.0268	0.0071/0.0061/0.0069	138.47/135.74/109.64
All small ϕ	0.0267/0.0267/0.0266	0.0178/0.0178/0.0179	0.0059/0.0060/0.0060	232.74/228.68/187.20
Big ϕ_τ , others small	0.0388/0.0390/0.0389	0.0181/0.0182/0.0182	0.0045/0.0044/0.0043	135.17/134.59/102.93

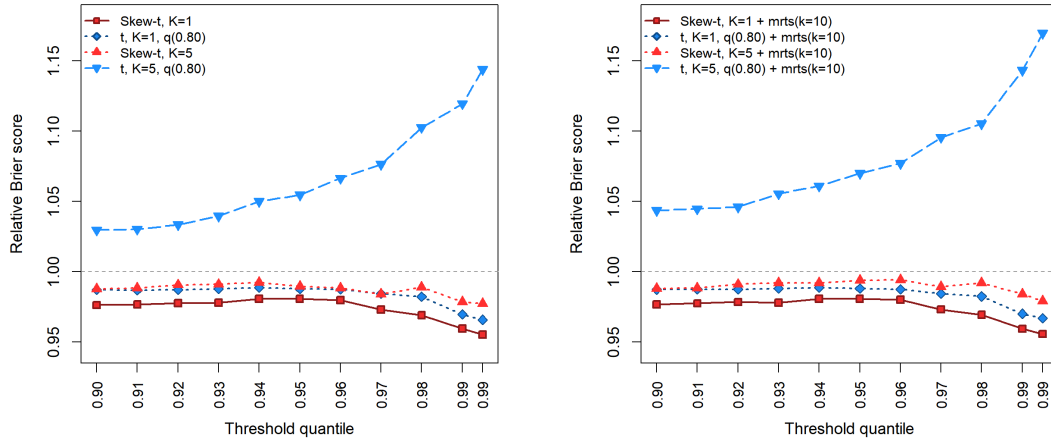
Table 1: Selected AR(2) simulation results under missing-at-location settings. Values are moved from `ar2_extend_detail_doc.md`; lower values indicate better performance.

For MRTS, the simulation evidence is more pattern-based than scenario-wise dominant: in setting 4 and setting 5, adding MRTS covariates can improve some upper-tail Brier or quantile scores, but the gains are not stable across all model families and quantile levels. Increasing the number of MRTS covariates also increases computation time, especially when $K_{\text{MRTS}} = 20$.

Figures ?? and ?? make this explicit by comparing methods 2–5 (skew- t and symmetric- t with $K = 1$ or $K = 5$ knots) at high threshold quantiles ($0.90 \leq q \leq 0.995$), with each figure showing two side-by-side panels: the left panel is the no-MRTS baseline and the right panel adds an MRTS basis of rank K_{MRTS} . All scores are normalized by the Gaussian baseline (method 1) computed without MRTS, so values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline. Setting 4 (skew- t with $K = 1$ knot, $\lambda = 3$) is shown for $K_{\text{MRTS}} \in \{10, 15, 20\}$ in Figure ??, and setting 5 (skew- t with $K = 5$ knots, $\lambda = 3$) is shown for $K_{\text{MRTS}} = 15$ in Figure ??.

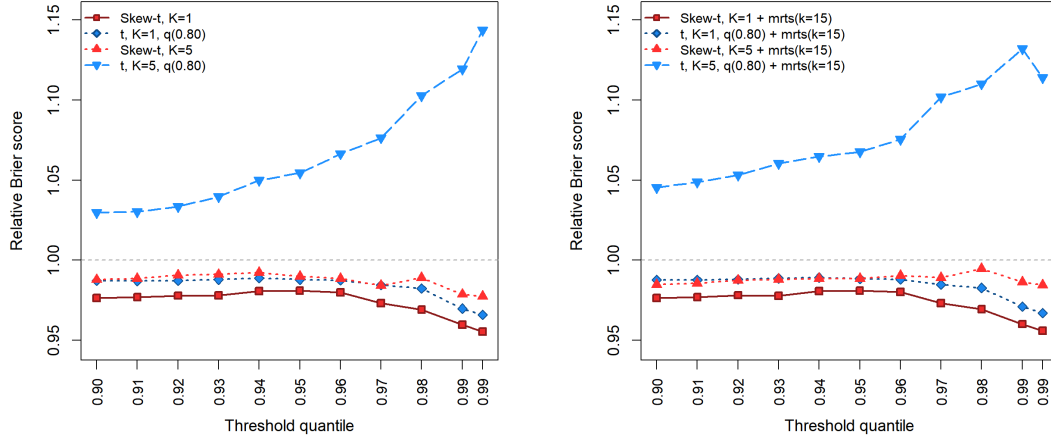
Two consistent patterns emerge. First, the data-generating knot count is the dominant axis: in setting 4 the one-knot skew- t method (method 2) is best at every quantile, and in setting 5 the five-knot skew- t method (method 4) is best at every quantile. Mis-specifying K on the under-side (fitting $K = 1$ to a $K = 5$ truth, as method 2 does

Data: Skew- t ($K=1, \lambda=3$)



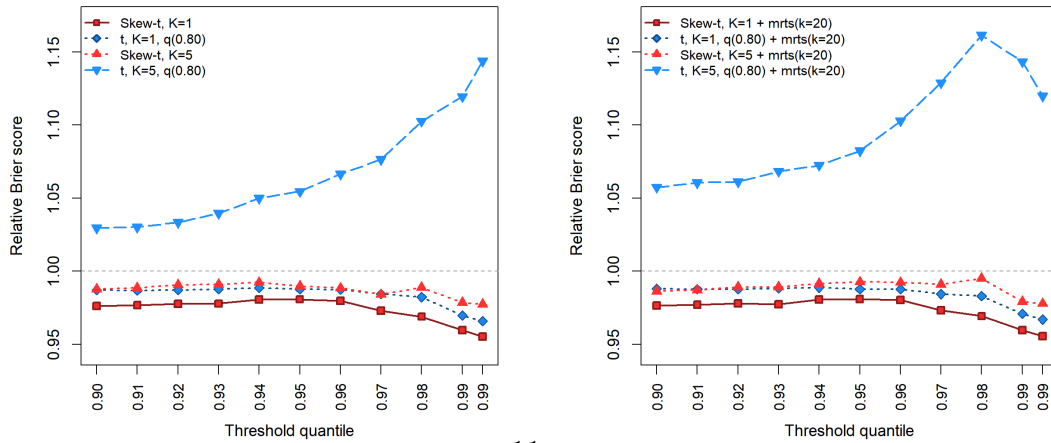
(a) $K_{\text{MRTS}} = 10$.

Data: Skew- t ($K=1, \lambda=3$)



(b) $K_{\text{MRTS}} = 15$.

Data: Skew- t ($K=1, \lambda=3$)



(c) $K_{\text{MRTS}} = 20$.

Figure 2: Setting 4 (data-generating process: skew- t with $K = 1$ knot, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline at threshold quantiles $0.90 <$

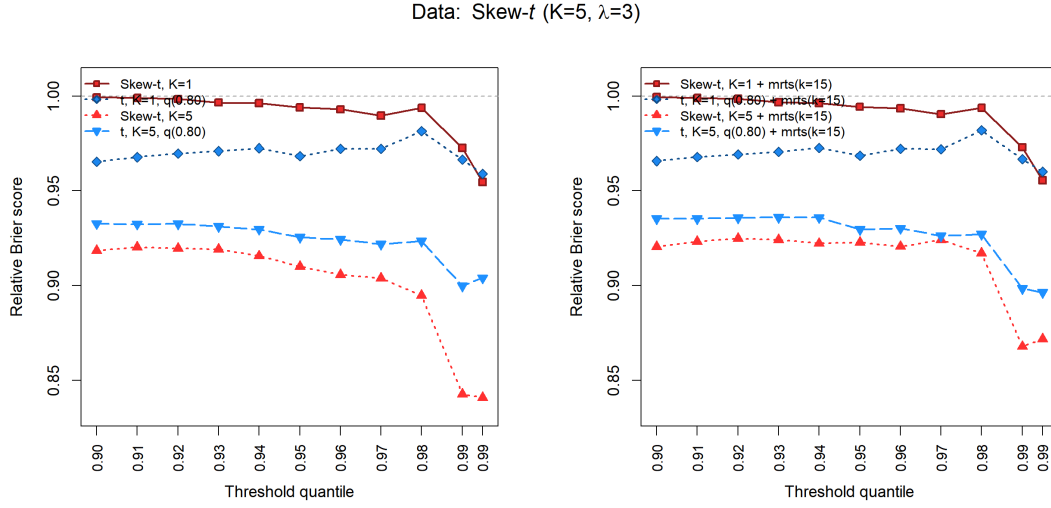


Figure 3: Setting 5 (data-generating process: skew- t with $K = 5$ knots, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline; left panel is the no-MRTS baseline, right panel adds an MRTS basis of rank $K_{\text{MRTS}} = 15$. The matching five-knot skew- t method (method 4) is decisively the best across all quantiles, with relative Brier scores between roughly 0.84 and 0.92. All four methods improve on the Gaussian baseline, in contrast with setting 4 where the over-specified method 5 was worse than Gaussian. The right panel mirrors the left panel closely, indicating that the MRTS basis adds little beyond what the spatial skew- t structure already captures.

in setting 5) gives only marginal gains over Gaussian, whereas mis-specifying K on the over-side (fitting $K = 5$ to a $K = 1$ truth, as method 4 does in setting 4) is competitive but not optimal. Second, the MRTS basis is essentially flat across the configurations studied here: for both settings and for every $K_{\text{MRTS}} \in \{10, 15, 20\}$, the right (with-MRTS) panel reproduces the left (without-MRTS) panel almost exactly. The MRTS basis therefore does not change the within-setting ranking, and its modest cost at $K_{\text{MRTS}} = 20$ is not repaid by tail-score improvement under these data-generating processes. Table ?? places AR(2) and MRTS side-by-side as proposed extensions in a common decision summary.

Extension		Simulation setup	Main empirical pattern	Practical implication
AR(2) time-varying coefficients		Missing-at-location experiments with AR(2), AR(1), and no-time-series fitted to multiple dependence regimes	AR(2) can improve selected mid-tail Brier scores, but runtime advantages are often with AR(1) or no-time-series; no uniform winner across regimes	Use AR(2) as a candidate temporal model, not as a default replacement
MRTS covariates		Benchmark simulation settings 4–5 with additional multi-resolution thin-plate spline covariates ($K_{\text{MRTS}} = 10, 15, 20$ or 15)	Improvements appear at selected tail levels, but average differences are small and not stable across quantiles/model classes; runtime increases with larger K_{MRTS}	Use MRTS selectively when nonstationarity is plausible and tail-score gains justify added cost

Table 2: Integrated simulation summary for the two proposed extensions.

Overall, the simulation study indicates that both extensions expand the usable model class, but neither extension uniformly dominates the Morris baseline. The AR(2) structure provides temporal flexibility, and MRTS covariates enrich spatial nonstationarity representation; however, their empirical benefits depend on data-generating conditions and target exceedance levels. Therefore, in the ozone analysis, AR(2) and MRTS should be treated as candidate options to be selected by held-out predictive scores.

Recommended additional experiments to strengthen logical completeness:

(1) Add a full AR(2) vs AR(1) vs NoTS experiment under the exact benchmark simulation design ($n_s = 144$, $n_t = 50$, same train/test split as Morris), so temporal-structure comparisons are directly comparable with the baseline five-model benchmark.

(2) Add a factorial AR(2)×MRTS experiment (e.g., AR(2) with $K_{\text{MRTS}} = 0, 10, 15, 20$) to test whether temporal flexibility and nonstationary covariates are complementary or redundant. This is currently missing, so the thesis cannot yet quantify interaction effects between the two proposed extensions.

(3) Add replicated uncertainty summaries (mean $\pm$ SD or median with interval) for all MRTS comparisons at each exceedance/quantile level, matching the AR(2) detailed tables, to improve comparability and statistical interpretability.

(4) Add sensitivity analysis for MCMC budget (short vs long chains) and report score stability per unit time, so computational cost comparisons between AR(2), AR(1), NoTS, and MRTS variants are on a fair basis.

4 Ozone data analysis

The real-data analysis applies the model comparison framework to daily maximum 8-hour ozone observations from July 2005. The response is the AQS monitor observation, and the main covariate is the corresponding CMAQ model output. Following the ozone analysis in Morris et al. (2017), the regression component uses $\mathbf{X}_t(\mathbf{s}) = [1, \text{CMAQ}_t(\mathbf{s})]^\top$, so the model is interpreted as a statistical calibration of CMAQ using site-level observations. The full comparison data set contains 800 monitoring sites observed over 31 days. We use two-fold cross validation, with 400 sites used for training and 400 sites used for validation in each fold.

The purpose of this analysis is not only to reproduce the Morris ozone comparison, but also to evaluate whether the AR(2) time-varying coefficient extension and the MRTS covariate extension are useful for real ozone prediction. A preliminary MERRA-2 analysis with 100 sites and 100 time points showed that AR(2), AR(1), and no-time-series models have very similar MAPE, Brier scores, and quantile scores. The ozone analysis therefore focuses on whether the enlarged model class can improve held-out tail prediction rather than assuming that the more flexible extension must be better.

4.1 Model comparisons

The candidate set contains approximately one hundred model settings. The baseline settings include Gaussian, skew- t , symmetric- t , and max-stable models, with different numbers of knots, censoring thresholds, and time-series choices. For the skew- t and symmetric- t models, the number of knots includes values such as $K = 1, 5, 6, 7, 8, 9, 10$, and 15. Censoring thresholds include no censoring, a moderate threshold such as $T = 50$, and a high threshold such as $T = 75$ ppb. The threshold 75 ppb is scientifically

important because it corresponds to the air-quality exceedance level emphasized in the original ozone application.

The extended candidate set adds two families. The first replaces the AR(1) time-varying coefficient structure with the AR(2) structure developed in this thesis. The second adds MRTS covariates to selected Morris-type settings, so that nonstationary spatial structure can enter through multiple-resolution thin-plate spline basis functions. These extensions are compared against the original Morris-type specifications using the same cross-validation splits, so the ranking reflects predictive performance rather than differences in train-test construction.

For each fitted model, posterior predictive samples at the validation sites are used to compute Brier scores and quantile scores. The Brier scores are evaluated at upper-tail empirical thresholds $q(0.90)$, $q(0.95)$, $q(0.98)$, $q(0.99)$, and $q(0.995)$, and quantile scores are evaluated at the same probability levels. Scores are averaged over validation sites, days, and folds. Lower scores indicate better prediction.

Each MCMC chain is run for 30000 iterations with 25000 burn-in iterations. As in the simulation study, convergence of prediction summaries is more important here than direct interpretation of partition-specific latent variables, because multiple-partition models can suffer from label switching. The model-selection criterion is therefore based on held-out prediction scores, not on a visual comparison of latent partition labels.

4.2 Results

The real-data results support the use of a broader model class than the original baseline alone. In the enlarged ozone comparison, the top two models for Brier and quantile scores at the upper-tail levels include AR(2) and MRTS variants. Tables ?? and ?? summarize the leading models in the primary 400-site training and 400-site validation comparison. At the 0.90 Brier threshold, the best model is an MRTS skew- t specification and the runner-up is the corresponding Morris skew- t specification, with nearly indistinguishable scores. At higher thresholds, the leading model changes across score levels: AR2 models lead the Brier comparisons at 0.95 and 0.995 and the quantile-score comparisons at 0.98, 0.99, and 0.995, whereas Morris skew- t or symmetric- t specifications remain strongest at some intermediate extreme thresholds.

The ranking is not concentrated on a single universal model. Across Brier scores, the strongest settings typically combine CMAQ, a time-series component, moderate to high thresholds, and a small to moderate number of knots. Across quantile scores, AR(2)

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	MRTS skew- t , $K = 1$, $T = 0$, AR(1), 10 MRTS covariates	0.04467	Morris skew- t , $K = 1$, $T = 0$, AR(1)	0.04470
0.9500	AR2 skew- t , $K = 7$, $T = 50$	0.02664	AR2 skew- t , $K = 10$, $T = 50$	0.02672
0.9800	Morris skew- t , $K = 6$, $T = 50$, AR(1)	0.01257	Morris skew- t , $K = 5$, $T = 50$, AR(1)	0.01257
0.9900	Morris symmetric- t , $K = 9$, $T =$ 75, AR(1)	0.00688	Morris skew- t , $K = 5$, $T = 50$	0.00690
0.9950	AR2 symmetric- t , $K = 15$, $T =$ 75	0.00384	Morris symmetric- t , $K = 9$, $T =$ 75, AR(1)	0.00385

Table 3: Top two Brier-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	MRTS skew- t , $K = 1$, $T = 0$, AR(1), 5 MRTS covariates	5.033	Morris skew- t , $K = 5$, $T = 0$	5.033
0.9500	Morris skew- t , $K = 7$, $T = 50$, AR(1)	2.988	AR2 skew- t , $K = 10$, $T = 0$	2.988
0.9800	AR2 skew- t , $K = 10$, $T = 0$	1.423	Morris skew- t , $K = 9$, $T = 0$	1.424
0.9900	AR2 skew- t , $K = 6$, $T = 50$	0.797	Morris skew- t , $K = 9$, $T = 50$, AR(1)	0.798
0.9950	AR2 skew- t , $K = 6$, $T = 50$	0.441	AR2 skew- t , $K = 8$, $T = 50$	0.442

Table 4: Top two quantile-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

settings appear frequently among the best-ranked models, especially for $q(0.95)$ and above. MRTS settings are more selective: they improve some tail exceedance scores, but they are not uniformly preferred across all scores or quantile levels.

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	204 (MRTS, 0.041)	55 (Morris's, 0.041)	205 (MRTS, 0.041)	70 (Morris's, 0.041)
0.9500	70 (Morris's, 0.020)	41 (Morris's, 0.020)	12 (Morris's, 0.020)	39 (Morris's, 0.020)
0.9800	204 (MRTS, 0.008)	117 (AR2, 0.008)	67 (Morris's, 0.008)	111 (AR2, 0.008)
0.9900	117 (AR2, 0.003)	15 (Morris's, 0.003)	70 (Morris's, 0.003)	61 (Morris's, 0.003)
0.9950	15 (Morris's, 0.001)	70 (Morris's, 0.001)	61 (Morris's, 0.001)	117 (AR2, 0.001)

Table 5: Validation selection result using 300 sites to predict 100 sites

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	54 (Morris's, 0.050)	105 (AR2, 0.050)	35 (Morris's, 0.050)	215 (MRTS, 0.050)
0.9500	117 (AR2, 0.030)	116 (AR2, 0.030)	114 (AR2, 0.030)	70 (Morris's, 0.030)
0.9800	215 (MRTS, 0.015)	205 (MRTS, 0.015)	111 (AR2, 0.015)	58 (Morris's, 0.015)
0.9900	58 (Morris's, 0.008)	112 (AR2, 0.008)	216 (AR2, 0.008)	67 (Morris's, 0.008)
0.9950	29 (Morris's, 0.004)	35 (Morris's, 0.004)	60 (Morris's, 0.004)	116 (AR2, 0.004)

Table 6: Validation selection result using 200 sites to predict 200 sites

The small-sample stability check gives a similar but not identical conclusion. The primary 400/400 split uses empirical thresholds 73.25, 79.75, 88.50, 94.75, and 100.52 for the 0.90, 0.95, 0.98, 0.99, and 0.995 Brier comparisons, respectively. Under this split, the top setting numbers across these thresholds are 204, 111, 58, 68, and 124. When the experiment is repeated with 300 training sites and 100 validation sites, the leading settings become 204, 70, 204, 117, and 15; with 200 training sites and 200 validation sites, they become 54, 117, 215, 58, and 29. Thus, the same broad model families remain competitive, but the exact best setting changes across thresholds and sample splits. This instability is expected because extreme-ozone validation uses relatively few exceedances, especially at the 0.99 and 0.995 levels. Consequently, the real-data analysis should not be read as identifying a single universally optimal tuning setting, but as evidence that AR(2) and MRTS versions are useful additions to the candidate pool when the target is held-out tail prediction.

Overall, the real-data analysis provides stronger practical motivation for the proposed extensions than the simulation study alone. The AR(2) extension gives additional

temporal flexibility and appears in the leading models for several tail quantile scores. The MRTS extension gives a way to enrich the spatial regression component and can improve selected exceedance scores. Therefore, the final modeling recommendation is not to replace the Morris model by a single proposed specification, but to include AR(2) and MRTS versions in the candidate pool and select among them using held-out Brier and quantile scores.

5 MERRA-2 data analysis

This preliminary analysis is conducted on MERRA-2 hourly surface-temperature data to assess whether temporal extensions provide practical predictive gains in a controlled medium-size setting. We use 100 spatial locations and 100 hourly time points, with longitude and latitude retained for each site.

5.1 Data preprocessing and model setup

We randomly split the sites into training and testing sets using seed 123, with 70% of sites used for model fitting and 30% reserved for validation. The regression design is specified as

so each time point has an intercept and first-order spatial trend terms. Because the covariates are not standardized in this experiment, we set `nu.upper` = 10 to stabilize posterior exploration and avoid unstable large- ν behavior during MCMC.

All three temporal specifications are fitted under the same core configuration (skew- t margin, Matérn covariance, $K = 5$ knots, and no censoring threshold) to isolate the effect of temporal structure. Following the real-data experiment setup, each chain is run for 20000 iterations with 10000 burn-in iterations, yielding 10000 retained posterior draws per model.

5.2 Results

Table ?? summarizes quantile-score (QS) and Brier-score (BS) comparisons for no-temporal, AR(1), and AR(2) specifications. The table shows a clear but subtle trade-off: the no-temporal model performs slightly better on the QS metrics and achieves the smallest BS at the most extreme level (0.99), whereas AR(2) attains the best BS at the 0.90 and 0.95 levels. Because the numerical differences are very small, none of the

three temporal structures dominates uniformly; instead, the preferred model depends on whether one emphasizes bulk accuracy, mid-tail exceedances, or the far tail.

The additional summary metrics reinforce this interpretation. MAPE is slightly lower for AR(2) (0.01058699), with AR(1) close behind, while the no-temporal model is somewhat worse on this bulk-accuracy measure. The 95% predictive-interval coverage is also nearly identical across models (approximately 0.912–0.916), which suggests that temporal dependence changes predictive sharpness more than calibration. These modest gains come at a cost: computation time rises from about 777.26 seconds for no-temporal to 1041.65 seconds for AR(1) and 1075.83 seconds for AR(2). Taken together, the results indicate that AR(2) is a reasonable option when slightly better mid-tail behavior is desired, but the improvement is incremental rather than decisive.

Model	QS90	QS95	QS99	BS90	BS95	BS99
No Temporal	4.714	2.522	0.570	0.041	0.018	0.008
AR(1)	4.716	2.522	0.570	0.040	0.016	0.009
AR(2)	4.717	2.522	0.571	0.040	0.016	0.009

Table 7: Comparison of temporal specifications in the MERRA-2 experiment. Smaller values indicate better predictive performance.

6 Discussion

References

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